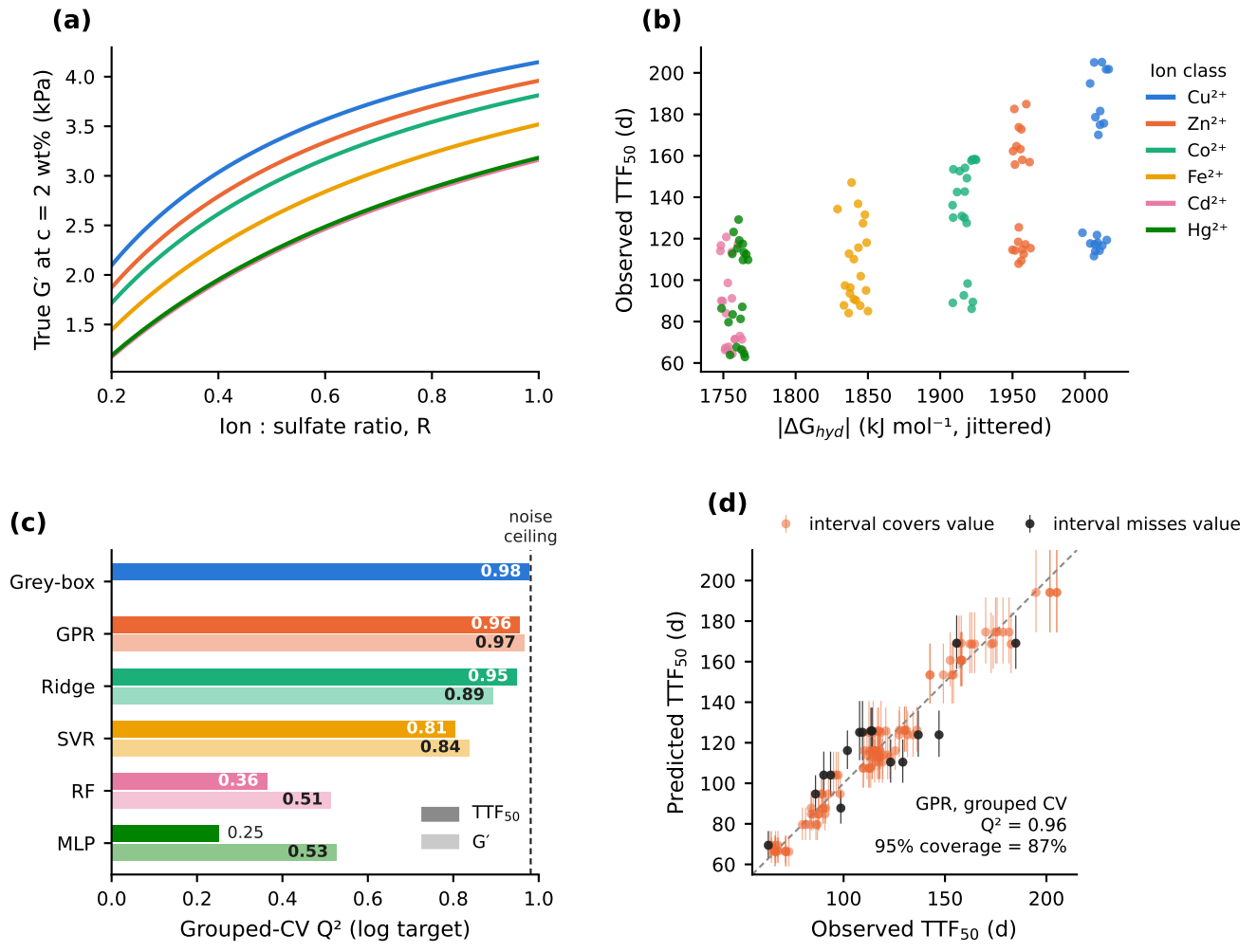
(a)
True G′ at c = 2 wt% (kPa)
4.0
3.5
3.0
2.5
2.0
1.5
0.2
0.4
0.6
0.8
1.0
Ion : sulfate ratio, R
(b)
Observed TTF50 (d)
200
180
160
140
120
100
80
60
1750
1800
1850
1900
1950
2000
|ΔGhyd| (kJ mol⁻¹, jittered)
Ion class
Cu²⁺
Zn²⁺
Co²⁺
Fe²⁺
Cd²⁺
Hg²⁺
(c)
noise ceiling
Grey-box
GPR
Ridge
SVR
RF
MLP
0.98
0.96
0.97
0.95
0.89
0.81
0.84
0.36
0.51
0.25
0.53
TTF50
G′
0.0
0.2
0.4
0.6
0.8
1.0
Grouped-CV Q² (log target)
(d)
interval covers value
interval misses value
Predicted TTF50 (d)
200
180
160
140
120
100
80
60
GPR, grouped CV
Q² = 0.96
95% coverage = 87%
100
150
200
Observed TTF50 (d)